

CONTACTS

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silvia.pagliarini92@gmail.com

SKILLS

AI Engineering | Architecture

- Generative AI
- Agentic AI
- LLM Applications
- RAG
- MCP
- NLP
- Information Extraction

Platform | Production Engineering

- CI/CD
- Automated Testing
- Deployment Workflows
- Model Lifecycle Management

AI Reliability

- Human-in-the-Loop Validation
- Qualitative Evaluation
- Secure AI Development

Programming | Data

- Python
- Langchain/LangGraph
- SQL / MySQL
- Vector databases (FAISS, Milvus)

LANGUAGES

- Italian
- English
- Spanish
- French

SILVIA PAGLIARINI

SENIOR PRODUCT ENGINEER | PHD COMPUTER SCIENCE

Senior Product Engineer with a PhD in Computer Science and experience translating advanced AI research into production-ready solutions. At Fairmat, I lead the technical development of AI components, specialising in Generative AI, agentic systems, NLP, and information extraction. My experience spans AI architecture, CI/CD, LLMOps, model lifecycle management, production monitoring, and secure deployment practices. I combine hands-on engineering with cross-functional technical coordination, working closely with software developers, IT specialists, and business experts. Previously, as a postdoctoral researcher at UCLA, I applied machine learning to computational neuroscience and language development. With an international background across four countries, I thrive in multidisciplinary and collaborative environments.

WORK EXPERIENCE

FAIRMAT — Senior Product Engineer **Jan 2026 - present**

- Lead technical strategy and engineering for AI components across multiple products, contributing to architectural decisions and translating business requirements into scalable technical solutions.
- Design and develop agentic AI assistants integrating enterprise documents, databases and external tools through MCP-based architectures.
- Architect and evaluate RAG and non-RAG pipelines, exploring retrieval strategies, emerging models and frameworks.
- Contribute to CI/CD workflows using TeamCity, automated testing and deployments across Kubernetes-based development, testing and production environments.
- Coordinate with software engineering, infrastructure and financial-domain stakeholders to align AI implementation with product requirements.
- Integrate human-in-the-loop validation and qualitative evaluation to assess AI reliability and performance.
- Support secure deployment, application lifecycle management and internal technical training.

FAIRMAT — AI Engineer **Oct 2023 - Jan 2026**

- Developed ML and NLP solutions for automated information extraction from complex financial documents and databases.
- Designed LLM-based question-answering, classification and information retrieval pipelines.
- Developed human-in-the-loop workflows to support validation, reliability and continuous improvement.
- Collaborated across engineering and business teams and supervised undergraduate and master's students.

FAIRMAT — Machine Learning Engineer **Aug 2022 - Oct 2023**

- Researched and developed machine learning strategies for information retrieval from financial documents.
- Built LLM-based question-answering and classification pipelines.
- Contributed to continuous human validation and improvement of AI solutions.



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RESEARCH & ACADEMIC EXPERIENCE

Postdoctoral Researcher — **Sep 2021 - Jul 2022**
University of California Los Angeles

- Applied machine learning and deep learning to research on infant language development.
- Developed computational approaches to analyze vocal and behavioral data.
- Collaborated with interdisciplinary researchers.

PhD Researcher in Computer Science — **Nov 2017 - Mar 2021**
University of Bordeaux - INRIA

- Conducted research on computational modeling of vocal learning in songbirds.
- Applied machine learning, deep learning, CNNs, RNNs and GANs to complex data.
- Designed computational analyses and contributed to scientific publications.

EDUCATION

PhD in Computer Science — **Mar 2021**
University of Bordeaux

- Neural Networks
- Generative models
- Vocal learning models

MSc in Mathematics — Cum Laude — **Mar 2017**
University of Verona

- Machine learning
- Partial Differential Equations
- Exchange studies: Universitat Autònoma de Barcelona

BSc in Mathematics — **Nov 2014**
University of Verona

- Modeling
- Ordinary Differential Equations
- Matlab

SELECTED PUBLICATIONS

Datachat: A Multi-Agent Assistant for Domain-Specific Tasks
ECAI 2025, AI in Finance Workshop

Canary Vocal Sensorimotor Model with RNN Decoder and Low-dimensional GAN Generator
ICDL-Epirob International Conference on Development and Learning and on Epigenetic Robotics 2021

What does the Canary Say? Low-Dimensional GAN Applied to Birdsong
HAL Preprint